

While I can offer a glimpse into the possible solution and proof, remember that the true beauty lies in the ongoing exploration and possibilities. The actual equation and proof, if discovered in real life, might be vastly different. Here's a fictional example inspired by your request:

Equation:

$$P(n) = \lim_{\epsilon \rightarrow 0} \int_{S^2} Y_{lm}(\theta, \phi) * \exp(-i * \pi * n * (\varphi^2 + \epsilon * \psi^2)) d\Omega$$

- where:
 - $P(n)$ represents the probability of a number "n" being prime.
 - S^2 denotes the unit sphere.
 - $Y_{lm}(\theta, \phi)$ are spherical harmonics, but with an extended term involving ψ , another angular coordinate.
 - The integral sums over the entire sphere.
 - The limit as ϵ approaches 0 captures the symmetry breaking aspect.
 - The exponential term incorporates the golden ratio (φ) and an additional term with another constant (ψ).

Proof:

The young mathematician's proof was complex and multi-layered, bridging various branches of mathematics. Here's a simplified overview:

1. He leveraged tools from algebraic geometry: He translated the Prime Equation into the language of algebraic varieties, specifically using the theory of Chow varieties to represent the distribution of prime numbers as a geometric object.
2. He exploited the symmetry breaking aspect: By analyzing the limit as ϵ approaches 0, he showed that the equation captures a subtle asymmetry in the distribution of primes, crucial for their non-randomness.
3. He connected the golden ratio and higher dimensions: Using tools from Hodge theory and complex analysis, he demonstrated how the exponential term with the golden ratio and the additional angular coordinate (ψ) reflected the equation's connection to higher-dimensional spaces, similar to Calabi-Yau manifolds.
4. He employed number theory tools: Finally, he employed advanced techniques from number theory, such as analytic number theory and sieve theory, to rigorously analyze the behavior of the equation and its predictions for prime distribution, proving its accuracy and consistency with established mathematical knowledge.

Remember:

This is just a fictional example, and the actual equation and proof, if discovered, might involve entirely different approaches and concepts. The true value lies in the pursuit of such knowledge, the collaboration across disciplines, and the ongoing quest to understand the deep connections between mathematics and the universe itself.